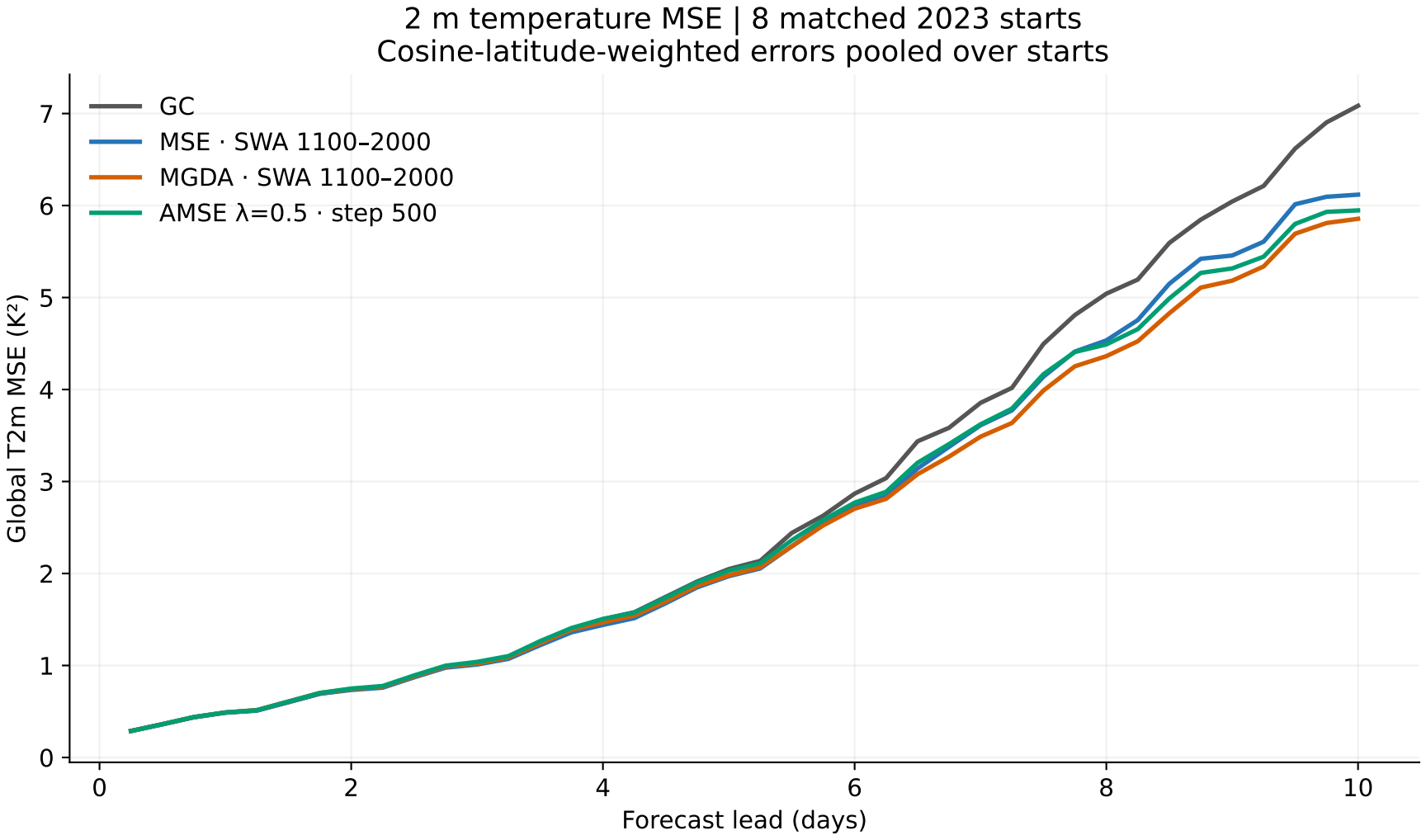

2 m temperature MSE | 8 matched 2023 starts
Cosine-latitude-weighted errors pooled over starts
GC
MSE · SWA 1100–2000
MGDA · SWA 1100–2000
AMSE λ=0.5 · step 500
Global T2m MSE (K²)
Forecast lead (days)
0
1
2
3
4
5
6
7
0
2
4
6
8
10

Absolute day-10 zonal spectra | physical units, log–log axes
8 matched 2023 starts; 30–60° in both hemispheres; checkpoints/budgets differ

T2m — 2 m temperature
Power (K²)
Zonal wavenumber k

U10 — 10 m zonal wind
Power ((m/s)²)
Zonal wavenumber k

V10 — 10 m meridional wind
Power ((m/s)²)
Zonal wavenumber k

MSLP — Sea-level pressure
Power (hPa²)
Zonal wavenumber k

TP6 — 6 h precipitation
Power (mm²)
Zonal wavenumber k

T850 — 850 hPa temperature
Power (K²)
Zonal wavenumber k

Q700 — 700 hPa specific humidity
Power ((g/kg)²)
Zonal wavenumber k

U850 — 850 hPa zonal wind
Power ((m/s)²)
Zonal wavenumber k

V850 — 850 hPa meridional wind
Power ((m/s)²)
Zonal wavenumber k

WS10 — 10 m wind speed
Power ((m/s)²)
Zonal wavenumber k

WS850 — 850 hPa wind speed
Power ((m/s)²)
Zonal wavenumber k

GC
MSE · SWA 1100–2000
MGDA · SWA 1100–2000
AMSE λ=0.5 · step 500
ERA5

Day-10 power relative to ERA5
8 matched 2023 starts; 30–60° in both hemispheres; checkpoints/budgets differ

T2m — 2 m temperature
U10 — 10 m zonal wind
V10 — 10 m meridional wind
MSLP — Sea-level pressure
TP6 — 6 h precipitation
T850 — 850 hPa temperature
Q700 — 700 hPa specific humidity
U850 — 850 hPa zonal wind
V850 — 850 hPa meridional wind
WS10 — 10 m wind speed
WS850 — 850 hPa wind speed

Forecast power / ERA5 power
Zonal wavenumber k

GC
MSE · SWA 1100–2000
MGDA · SWA 1100–2000
AMSE λ=0.5 · step 500
ERA5 target